

MOHAMED ABU BASITH

Generative AI Engineer | Software Engineer

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📍 Perungudi, Chennai

PROFESSIONAL SUMMARY

Results-driven Generative AI Engineer and Software Engineer with 4+ years of industry experience. Since January 2025, leading production-grade Gen AI projects including a real-time voice-to-voice conversational AI agent using LangChain, LiveKit, and OpenRouter, and an enterprise Knowledge Base platform featuring 7 RAG pipeline variants (Standard RAG, HyDE, Hybrid, Graph RAG) built on Kubeflow. Proficient in LLM APIs (OpenAI, Anthropic), vector databases, prompt engineering, MCP configurations, and AI agent development. Strong mobile engineering foundation in Android, Flutter, and React Native.

TECHNICAL SKILLS

▪ Large Language Models (LLMs)	▪ Prompt Engineering
▪ RAG (Retrieval Augmented Generation)	▪ AI Agents & Agentic AI
▪ LangChain	▪ LlamaIndex
▪ OpenAI API / Anthropic API	▪ OpenRouter (Multi-LLM Provider)
▪ Vector Databases (ChromaDB / FAISS)	▪ Embeddings & Semantic Search
▪ Knowledge Base Systems	▪ Graph RAG / HyDE / Hybrid RAG
▪ Conversational AI (Voice + Text)	▪ LiveKit (Real-Time Voice Agent)
▪ MCP (Model Context Protocol)	▪ Multi-Agent Systems
▪ Kubeflow / MLOps Pipelines	▪ Airbyte (Data Connectors)
▪ Web & API Scraping (Cron Jobs)	▪ Document Verification (Gen AI)
▪ Python	▪ JavaScript & TypeScript (TSX)
▪ Node.js	▪ Git / Jira
▪ Android / Flutter / React Native	▪ Firebase / REST APIs

PROFESSIONAL EXPERIENCE

Generative AI Engineer / Software Engineer | Grootan Technologies Pvt Ltd

Jan 2022 – Present

1st Floor 87, Bhaggyam Yukti, Industrial Estate, Perungudi, Chennai, Tamil Nadu 600096

Project 1: Conversational AI Agent – Voice & Text (2025)

- Built a production-grade real-time voice-to-voice AI agent using LiveKit for WebRTC audio streaming and low-latency communication.
- Integrated LangChain for LLM orchestration and OpenRouter as the multi-provider LLM configuration layer, enabling dynamic model switching across OpenAI, Anthropic, and other providers.
- Implemented MCP (Model Context Protocol) to connect the agent with external tools, APIs, and enterprise services for real-time data access.
- Designed vector database-backed knowledge base grounding with semantic search and embeddings for context-aware, hallucination-reduced responses.
- Built database connectivity layer for persistent session management and external resource access by the AI agent.

- Delivered a dual-mode (voice + text) conversational agent with prompt engineering-optimized system prompts for production reliability.

Project 2: Enterprise Knowledge Base Platform – 7-Pipeline RAG Architecture (2025)

- Architected a product-grade enterprise Knowledge Base platform using a multi-pipeline RAG concept with 7 retrieval strategies for diverse enterprise use cases.
- Implemented pipeline variants: Standard RAG, HyDE (Hypothetical Document Embeddings), Hybrid RAG, Graph RAG, and custom pipelines — using LlamaIndex and LangChain as the core orchestration frameworks.
- Integrated vector databases (ChromaDB, FAISS) with embedding models for high-accuracy semantic search and document retrieval.
- Built external source connectors using Airbyte for structured data ingestion from third-party systems and databases.
- Developed automated web scraping and API scraping modules with Cron job scheduling for continuous, up-to-date knowledge base population.
- Deployed and orchestrated the full multi-pipeline infrastructure on Kubeflow for scalable MLOps and enterprise-grade pipeline management.

Project 3: Gen AI Document Verification POC (2025) – Saudi Client

- Developed a POC for automated document verification using Generative AI and multimodal LLM capabilities for a Saudi-based enterprise client.
- Applied prompt engineering and LLM API integration (OpenAI) to extract, validate, and cross-reference structured and unstructured document data.

Mobile Application Development (2022 – 2024)

- Designed, developed, and maintained Android, Flutter, and React Native applications across multiple client projects.
- Integrated Android native functionalities into a React Native wrapper SDK for cross-platform compatibility.
- Delivered Firebase-backed mobile applications with real-time database, authentication, and REST API integration.

Software Engineer – Intern | Grootan Technologies Pvt Ltd

Jul 2021 – Dec 2021

1st Floor 87, Bhaggyam Yukti, Industrial Estate, Perungudi, Chennai, Tamil Nadu 600096

- Designed and developed mobile applications for phones and tablets with clean UI and solid performance.
- Integrated backend APIs and database response servers for server-side logic.
- Built features using Firebase Authentication and Firebase Real-time Database.

EDUCATION

Master of Business Administration (MBA) – Marketing Management

2022 – 2024

Alagappa University, Karaikudi

First Class – December 2023 | Reg. No.: 2022013600080

Bachelor of Computer Application (BCA)

2017 – 2020

Jamal Mohamed College, Tiruchirappalli

First Class | Roll No.: 17US9753

CERTIFICATION

Ethical Hacking

INTERESTS & HOBBIES

Playing Badminton